Stoichiometry Involving Solutions Worksheet

1. Calculate the number of mL of 2.00 M HNO_3 solution required to react with 216 grams of Ag according to the equation.

$$3\ Ag_{(s)} + 4\ HNO_{3(aq)} \text{ ---------> } 3\ AgNO_{3(aq)} + NO_{(g)} + 2\ H_2O_{(l)}$$

2. Calculate in mL the volume of 0.500 M NaOH required to react with 3.0 grams of acetic acid. The equation is:

$$NaOH_{(aq)} + HC_2H_3O_{2(aq)} \text{ -------> } NaC_2H_3O_{2(aq)} + H_2O_{(l)}$$

3. Calculate the number of grams of AgCl formed when 0.200 L of 0.200 M $AgNO_3$ reacts with an excess of $CaCl_2$. The equation is:

$$2\ AgNO_{3(aq)} + CaCl_{2(aq)} \text{ -------> } 2\ AgCl_{(s)} + Ca(NO_3)_{2(aq)}$$

4. Calculate the mass of AgCl formed when an excess of 0.100 M solution of NaCl is added to 0.100 L of 0.200 M $AgNO_3$.

5. To neutralize the acid in 10.0 mL of 18.0 M H_2SO_4 that was accidentally spilled on a laboratory bench top, solid sodium bicarbonate was used. The container of sodium bicarbonate was known to weigh 155.0 g before this use and out of curiosity its mass was measured as 144.5 g afterwards. The reaction that neutralizes sulphuric acid this way is as follows.

$$H_2SO_4 + 2\ NaHCO_3 \text{ --------> } Na_2SO_4 + 2\ CO_2 + 2\ H_2O$$

Was sufficient sodium bicarbonate used? Calculate the limiting reactant and the maximum yield in grams of sodium sulphate.